

Abstract

This experiment presents the design and simulation of a three-digit digital password lock system using basic digital logic components. The system allows a predefined 3-digit password to be stored and compares it with user input. Logic gates, flip-flops, multiplexers, and comparators are used to implement password setting, user input verification, and door control. The circuit was simulated in Multisim, and the results show that the door opens only when the entered password exactly matches the stored password, while invalid inputs are rejected.

Introduction

Digital password lock systems are widely used in security applications such as door access control, lockers, and embedded systems. These systems improve security by allowing access only when a correct password is entered. In this project, a three-digit password lock is designed using digital logic circuits without using a microcontroller. The objective of this project is to understand how combinational and sequential logic components can be integrated to implement a basic security system and how binary-coded decimal (BCD) logic can be used to restrict valid inputs.

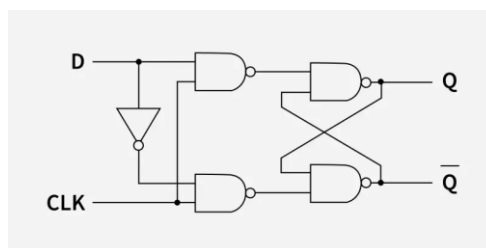
Theory

A digital password lock operates by comparing stored data with user-entered data. In this experiment, each digit of the password is represented using **4-bit BCD (Binary Coded Decimal)**. Since BCD supports only decimal values from 0 to 9, additional logic is required to prevent invalid values (10–15).

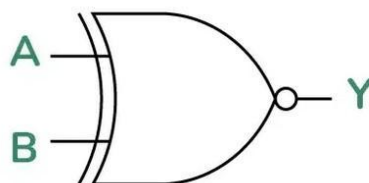
D Flip-Flops (D-FF): Used to store each bit of the password. When the clock signal is low, the flip flop holds its current state and ignores the D input. When the clock signal is high, the flip flop samples and stores D input.

If $D = 0$ then Q will be 0.

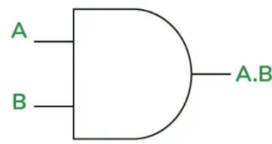
If $D = 1$ then Q will be 1.



XNOR Gates: Used to compare stored password bits with user input bits. In XNOR gate the Output is high (1) when both inputs are same (either both 0 or both 1), and low (0) when the inputs are different.



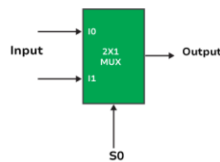
AND Gates: Used to ensure that all bits match simultaneously. If all the inputs are 1, Output will be 1 if either input is zero, Output will be 0.



Multiplexers (MUX): Used to select between password-setting mode and user-input mode. The 2x1 mux has two input lines, one output line, and a single selection line. It has various applications in digital systems such as in microprocessor it is used to select between two different data sources or between two different instructions.

- When S_0 is 0(low), the I_0 is selected
- when S_0 is 1(High), I_1 is selected

2:1 Multiplexer



Truth Table

S_0	I_0	I_1	Y
0	0	X	0
0	1	X	1
1	X	0	0
1	X	1	1

Methodology

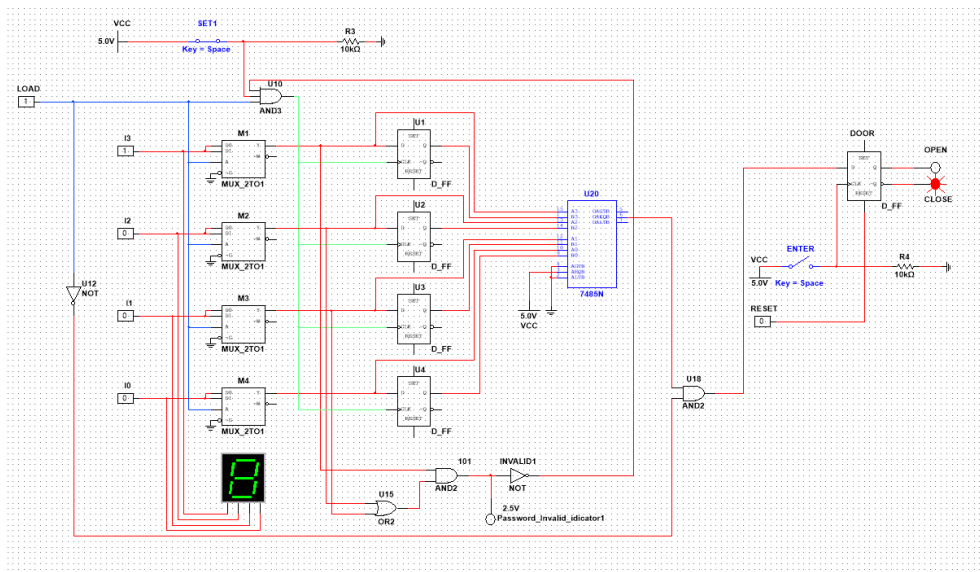
A total of **three digits** is used for the password, each represented by **4 switches** (BCD format). During password setup mode, the password bits are stored in D flip-flops using a load control signal. During user input mode, the entered digits are compared with the stored password using comparator. BCD validation logic is applied to ensure that only digits from 0 to 9 are accepted. The outputs of all comparators are combined using AND gates. If all digits match and the input is valid, the door control output becomes HIGH. LEDs are used to indicate door open (green) and door closed (red). The entire circuit was designed and simulated using **Multisim**.

Result

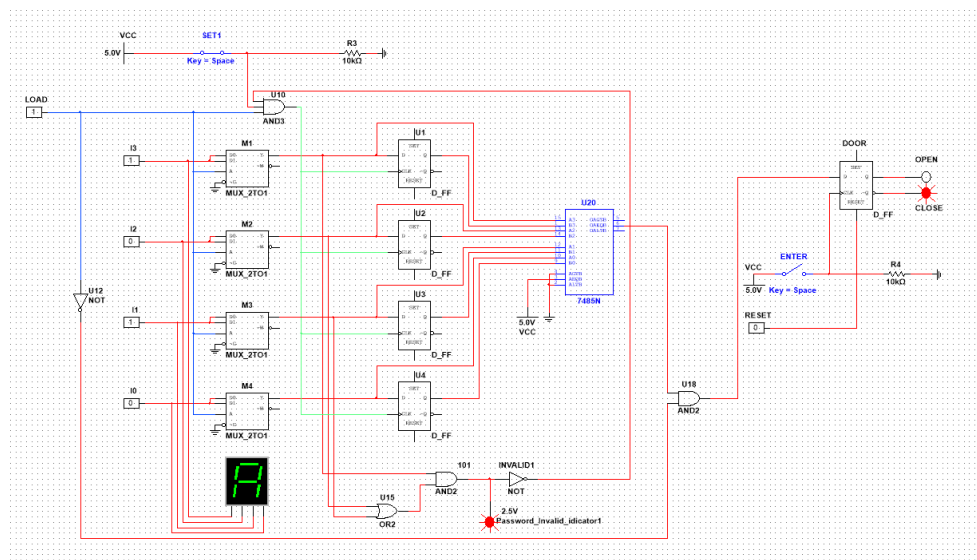
The simulation results confirm the correct operation of the 3-digit password lock system:

When the correct 3-digit password is entered, the green LED turns ON, indicating that the door is open. When an incorrect password is entered, the red LED remains ON, indicating the door is closed. Inputs from 10 to 15 are successfully rejected by the BCD validation logic, indicating the red light. The door does not open if the load or enable condition is not satisfied.

Simulation:



When Load is 1 input 8(1000) is set as password, DOOR remain close



When Load is 1, input 10(1010) is not set as password, it will indicate the Password_Invalid_indicator then manual D flip-flop clock will not active, DOOR remain close

